

RAHUL MITRA

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EDUCATION

National University of Singapore

Bachelor of Engineering, Industrial Systems Engineering

Singapore
Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.

EXPERIENCE

STMicroelectronics

Operations Research (NUS System Design Project), Contract

Singapore
Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Robotics Vision Engineer (Blended AI+X)

Singapore
Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Abbott Laboratories

Operational AI Systems & Data Engineer (Contract)

Singapore
Jun 2026 – Present

- Build an air-gapped process-intelligence platform for globally distributed spray-drying plants, spanning GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted quantized Gemma model.

Abbott Laboratories

Data Analytics and Project/Product Management Intern

Singapore
Jan 2026 – Jun 2026

- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of unseen raw data with zero execution failures.
- Productionized a team-built APC simulator for live manufacturing and engineer training, packaging it in Docker and deploying it to Azure App Service.

People's Association

Software Engineer (Citizen Developer)

Singapore
Sep 2025 – Present

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.

Republic of Singapore Navy

Base Support Assistant

Singapore
Jan 2022 – Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.

PROJECTS

AsyncDDGS — Open-Source Async Search Client (PyPI)

2024 – Present

- Authored and maintain an asyncio-first, aiohttp-based DuckDuckGo client on PyPI with pytest CI/CD, adopted across AI tooling, Discord bots, and search projects.

Interactive Portfolio Infrastructure

2026

- Built an AI-assisted portfolio in React, TypeScript, Three.js, and Matter.js with profile switching, searchable field notes, analytics, and a 3D project world.

OnTheSpectrum — Codex & Blender MCP 3D Pipeline

May 2026

- Built a local-first Blender-to-Three.js asset pipeline using Codex and Blender MCP, generating GLB assets, metadata, previews, playable-world QA, and Remotion video workflows.

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre

Elderly Care Volunteer

Singapore
Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

SKILLS AND CERTIFICATIONS

Technical: Python, TypeScript, React, Next.js, Three.js, FastAPI, aiohttp/asyncio, PyTorch, ONNX, Docker, CI/CD, Azure; 3D CV: projective & epipolar geometry, absolute pose, SfM, bundle adjustment, multi-view stereo

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)